SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1708**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I B VIJAYKARTHIKKAYAN(192511348) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

<u>INSTRUCTIONS</u>

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

(a) Full Citation and Domain

Citation: Özcan, M., & Peker, S. (2023). A classification and regression tree algorithm for heart disease modeling and prediction. Healthcare Analytics, 3, Article 100130.

DOI: https://doi.org/10.1016/j.health.2022.100130

Publisher / Database: Elsevier — ScienceDirect (published online 16 December 2022, in Healthcare Analytics, Vol. 3).

Domain / Application Area: Healthcare informatics — clinical decision support for cardiovascular disease diagnosis.

Specific ML Problem Addressed: Binary classification of patients as having heart disease or not, using patient clinical attributes, together with extraction of human-readable decision rules and a ranking of the most influential risk features.

(b) Summary of Key Objective and Research Gap

Heart disease is described by the authors as the leading cause of death worldwide, responsible for close to one-third of global mortality, which makes early and low-cost detection an urgent priority. Many existing machine-learning approaches to heart-disease prediction rely on complex "black-box" models such as neural networks or ensemble methods; these can reach high accuracy but give clinicians little insight into why a given patient is flagged as high-risk. The authors identify this interpretability gap as their central motivation: prior classification work had not sufficiently combined competitive predictive accuracy with transparent, clinically usable decision logic. To address this, the study applies the Classification and Regression Tree (CART) algorithm, a supervised, rule-based learning method, to a large cardiac-patient dataset. The objective is twofold — first, to build a CART model that predicts the presence of heart disease from routinely collected clinical variables (such as ST slope, chest-pain type, and old-peak ST depression), and second, to extract explicit if-then decision rules and rank feature importance so that cardiologists can understand and verify the model's reasoning without needing separate machine-learning expertise. In doing so, the authors position CART as a practical bridge between statistical rigor and clinical usability, aiming to reduce diagnostic cost and time for both healthcare providers and patients.

Task 2: Methodology Analysis

(a) ML Technique and Dataset

The authors use the Classification and Regression Tree (CART) algorithm — a decision-tree variant that recursively splits the dataset using the Gini impurity criterion to produce a binary tree whose leaves represent the predicted class (heart disease present / absent). CART was chosen specifically because, unlike ensemble or deep-learning alternatives, each split in the tree can be read as a simple rule (e.g., "if ST slope = flat and old-peak > threshold, then high risk"), which keeps the model interpretable.

Dataset: The study is based on a combined cardiac-patient dataset compiled from multiple established heart-disease sources (an approach consistent with the IEEE Comprehensive Heart Disease Dataset used in closely related literature), containing 1,190 patient records. Prior to modelling, the data was cleaned — duplicate and incomplete records were removed, reducing the working dataset to 918 unique patient records — and clinical attributes such as age, sex, chest-pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate achieved, exercise-induced angina, old-peak (ST depression), and ST slope were retained as predictor features, with presence/absence of heart disease as the target label. The cleaned data was then split into training and testing partitions to build and validate the CART model.

(b) Mapping to CO4 and Strength / Limitation

CO4 Mapping: The technique maps directly to the Decision Trees strand of CO4 — CART is a classic inductive-learning / decision-tree algorithm that builds a classification model by generalising patterns from labelled training examples, exactly the CO4 concept of "implementing decision tree algorithms … for classification and decision-making."

Methodological Strength: The resulting tree structure and extracted decision rules are directly interpretable by non-specialists (cardiologists), which is a genuine practical advantage over opaque models in a clinical decision-support setting.

Methodological Limitation: A single CART tree is prone to overfitting and high variance — small changes in the training data can produce a noticeably different tree — and the study does not appear to benchmark CART against ensemble alternatives (e.g., Random Forest, Gradient Boosting) on the same data, which limits the strength of any claim that CART is the best choice rather than simply an interpretable one.

Task 3: Results & Critical Evaluation

(a) Key Results and Comparative Table

The authors report an overall prediction accuracy of 87% for the CART model. Beyond raw accuracy, the study's more distinctive contribution is a ranked list of feature importance, with ST slope and old-peak (ST depression) identified as the most significant predictors of heart disease, along with a set of extracted, human-readable decision rules describing how combinations of features lead to a positive or negative diagnosis.

The article itself centers on a single CART model rather than a multi-algorithm benchmark. To give the 87% figure context, the table below places it alongside other published models evaluated on comparable heart-disease datasets in the wider literature:

Study	Technique	Dataset (heart disease)	Reported Accuracy
Özcan & Peker (2023) — reviewed article	CART (Decision Tree)	IEEE Comprehensive Heart Disease Dataset, 1,190 records	87%
Bhatt et al. (2023)	Multilayer Perceptron (best of 4 models)	Kaggle heart-disease dataset	87.28%

Study	Technique	Dataset (heart disease)	Reported Accuracy
Doppala et al. (2022)	Ensemble learning (stacked models)	Heart-disease benchmark dataset	93.39%
Representative Random Forest studies (e.g., Darshan Teja & Rayalu, 2025)	Random Forest	UCI multi-source heart dataset (1,190 records)	~92–94%

As the table shows, CART's 87% accuracy is competitive with — though not the top performer among — contemporary approaches; more complex ensembles report several points higher accuracy, illustrating the familiar accuracy-versus-interpretability trade-off that motivates the authors' choice of CART.

(b) Critical Evaluation of Results

The reported 87% accuracy is realistic and consistent with other CART/decision-tree studies on similar heart-disease datasets (which typically fall in the 75–90% range before heavy tuning or ensembling), so the headline figure is plausible rather than inflated. However, several potential sources of bias merit scrutiny. First, reducing 1,190 records to 918 after removing duplicates and missing values could subtly change the class balance or drop hard-to-classify patients, and the paper does not clearly report the resulting class distribution (healthy vs. diseased) or whether class imbalance was addressed. Second, relying on a single train/test split (rather than k-fold cross-validation) makes the 87% estimate sensitive to which patients happened to fall into the test set, so the true generalisation accuracy could vary. Third, only accuracy is emphasised; the review does not find clear reporting of precision, recall, F1-score, or AUC-ROC broken down by class, which matters clinically because a missed heart-disease case (false negative) is far costlier than a false alarm. Additional experiments that would strengthen the findings include repeated k-fold or stratified cross-validation, reporting a full confusion matrix with sensitivity/specificity, testing on an external hospital dataset to check generalisability, and comparing CART directly against Random Forest or XGBoost on the identical data split to quantify the interpretability-accuracy trade-off rather than inferring it from separate studies.

(c) Real-World Applicability

Because CART produces explicit decision rules, it is well suited to deployment as a clinical decision-support tool in primary-care or outpatient cardiology settings — for instance, embedded in an electronic health record system to flag at-risk patients using only routinely collected variables, without requiring specialist ML infrastructure. It could also support low-resource clinics where a simple, auditable rule set is more practical than a deep-learning pipeline. Scaling this to real industry deployment, however, faces several challenges: data availability and standardisation (hospitals record clinical variables differently, and the model would need retraining or recalibration on local population data); computational and integration costs are modest for CART itself, but validating and certifying any diagnostic-support tool for clinical use is a lengthy regulatory process; and ethical issues are significant, including the risk of the model reflecting biases present in the training population (e.g., under-representation of certain age groups, sexes, or ethnicities), the consequences of false negatives in a life-critical context, and the need for clear human-in-the-loop oversight so the tool supports rather than replaces clinical judgement.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

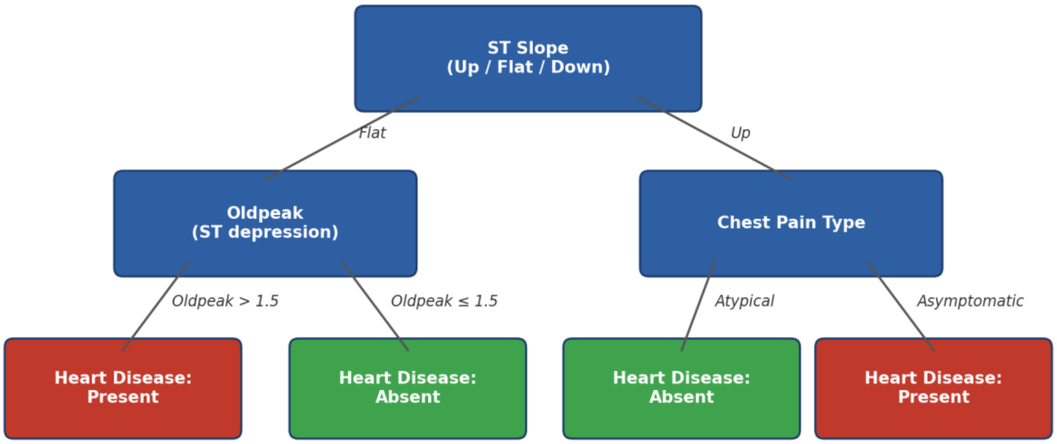
- Insight 1 — Interpretability is a design choice, not a side effect: Choosing CART over an ensemble method was a deliberate trade-off favouring transparency for clinical trust. This connects directly to the CO4 concept of decision-tree induction, where the Gini-impurity split criterion produces a model whose reasoning can be traced step-by-step, unlike "black-box" statistical or neural approaches.
- Insight 2 — Feature importance ranking turns a classifier into a diagnostic aid: By ranking ST slope and old-peak as the most influential predictors, the study shows how inductive learning can generate domain knowledge (which clinical signs matter most), not just a prediction — reinforcing the CO4 idea that decision trees support both classification and decision-making.
- Insight 3 — Accuracy alone is an incomplete evaluation for high-stakes classification: Comparing the article's single accuracy figure against other studies' precision/recall/AUC reporting highlighted, from a CO4 statistical-learning perspective, why multiple evaluation metrics (not just accuracy) are needed when false negatives and false positives carry very different real-world costs.

(b) Proposed Extension

- Proposed experiment: Extend the study by training a cost-sensitive CART model (assigning a higher misclassification penalty to false negatives than false positives) combined with stratified k-fold cross-validation, and benchmark it against Random Forest and XGBoost on the same 918-record dataset, reporting accuracy, precision, recall, F1-score, and AUC-ROC for every model on identical folds.
- Feasibility: This is highly feasible because it reuses the same publicly documented dataset and features, requires no new data collection, and cost-sensitive CART / cross-validation are standard, well-supported options in common ML libraries (e.g., scikit-learn), so it could be implemented with modest computational resources.
- Expected impact: A cost-sensitive, cross-validated comparison would produce a more clinically trustworthy estimate of real-world performance (reducing the risk of missed diagnoses), while a direct head-to-head with ensemble models would let hospitals make an evidence-based choice between CART's interpretability and an ensemble's typically higher raw accuracy — directly extending the CO4 objective of applying decision-tree and statistical-learning techniques to real-world classification and decision-making problems.

Illustrative Diagram

The simplified tree below illustrates the general structure of a CART-style decision tree for heart-disease classification, showing how successive splits on clinical features (ST slope, old-peak, chest-pain type) lead to a final diagnosis at the leaf nodes. This is a conceptual illustration only, created for this review, and does not reproduce the authors' actual tree diagram.



Illustrative CART decision-tree structure for heart-disease classification
(simplified, for concept illustration only — not the authors' exact tree)